

ACADEMIC REPORT

Project N°3

PGM IMAGE PROCESSING

Computer Science Department

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Abstract

This report presents a comprehensive implementation of image processing functionalities for Portable Gray Map (PGM) file format using the C programming language. The project encompasses fundamental operations including image initialization, memory management, pixel manipulation, geometric transformations, and steganographic techniques. Through systematic development and rigorous testing, this implementation demonstrates the practical application of data structures, dynamic memory allocation, and algorithmic problem-solving in the context of digital image processing. The findings reveal both the capabilities and limitations of PGM-based steganography while providing insights into efficient low-level image manipulation techniques.

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1 Introduction

Digital image processing represents a fundamental domain within computer science, bridging theoretical algorithmic concepts with practical applications in multimedia systems, computer vision, and data analysis. The Portable Gray Map (PGM) format, as defined by the Netpbm project, provides an accessible yet powerful platform for understanding core image manipulation principles without the complexity of compressed formats like JPEG or PNG.

This project investigates the implementation of essential image processing operations through the development of a C-based library capable of handling PGM files. The ASCII variant of PGM (P2 format) was selected for this implementation due to its human-readable structure, which facilitates debugging and educational exploration. Each pixel in a PGM image is represented by a single grayscale intensity value ranging from 0 (black) to a specified maximum value (typically 255), making it particularly suitable for fundamental image processing experiments.

1.1 Research Objectives

The primary objectives of this research project are threefold. First, to design and implement a robust data structure for representing grayscale images in memory, incorporating appropriate metadata such as dimensions and maximum intensity values. Second, to develop a comprehensive suite of image manipulation functions including geometric transformations, pixel-level operations, and intensity inversions. Third, to explore steganographic techniques by investigating methods for embedding and extracting textual

information within image data while maintaining visual fidelity.

1.2 Scope and Significance

Understanding image processing at a low level provides critical insights into memory management, algorithmic efficiency, and data structure design—skills that transfer to broader software engineering contexts. Furthermore, the steganographic component of this project introduces concepts relevant to information security, digital watermarking, and covert communication channels. By working directly with pixel arrays and implementing algorithms from first principles, this project cultivates a deeper appreciation for the computational foundations underlying modern image processing frameworks and libraries.

2 Methodology and Implementation

The implementation follows a structured approach, beginning with the definition of appropriate data structures and progressing through increasingly complex operations. This section details the architectural decisions, algorithmic approaches, and technical considerations that shaped the final implementation.

2.1 Data Structure Design

At the core of the implementation lies a structure designed to encapsulate all relevant image properties. The structure maintains four essential fields: the number of rows, the number of columns, the maximum grayscale value, and a dynamically allocated two-dimensional array storing pixel intensities. This design decision enables efficient random access to pixels while maintaining

clear separation between image metadata and pixel data.

```
typedef struct {  
    int rows;  
    int cols;  
    int max_gray;  
    int** pixels;  
} PGM_Image;
```

Dynamic memory allocation was employed for the pixel array to accommodate images of varying dimensions. This approach requires careful attention to memory management, with corresponding deallocation functions implemented to prevent memory leaks. The allocation strategy follows a two-step process: first allocating an array of row pointers, then allocating individual row arrays. This structure provides both the flexibility of dynamic sizing and the convenience of two-dimensional array indexing.

2.2 Core Operations Implementation

Several fundamental operations were implemented to provide comprehensive image manipulation capabilities. The initialization function establishes the image structure with specified dimensions and allocates the necessary memory. A complementary deep copy function creates independent duplicates of images, essential for non-destructive editing workflows. Accessor and mutator functions provide controlled access to individual pixel values with boundary checking to prevent buffer overflows.

The image negation operation demonstrates a straightforward yet practical transformation. By subtracting each pixel value from the maximum gray level, the function inverts the intensity distribution, producing a photographic negative effect. This operation illustrates the principle of intensity transformations and serves as a foundation for more complex histogram-based manipulations.

2.3 Geometric Transformations

Rotation operations present particular algorithmic challenges due to the need to map source pixels to destination coordinates while handling boundary conditions. The implementation supports rotation by arbitrary angles specified in radians. For each destination pixel, the algorithm computes the corresponding source coordinates using inverse rotation matrices. This reverse mapping approach prevents gaps in the output image that could arise from forward mapping strategies.

The rotation algorithm incorporates several important considerations. First, it maintains the original image dimensions, which necessitates clipping portions that rotate out of bounds. Second, it employs nearest-neighbor interpolation for determining pixel values at non-integer coordinates, accepting the trade-off between computational efficiency and potential aliasing artifacts. Third, it initializes all destination pixels to zero before mapping, ensuring that any unmapped regions appear black rather than containing random memory values.

Specific angle implementations (90°, 180°, 270°) were optimized with direct

coordinate transformations rather than trigonometric calculations. These special cases execute significantly faster and avoid floating-point precision issues while producing pixel-perfect results. The general rotation function serves cases requiring arbitrary angles, such as fine-tuned alignment corrections or creative effects.

3 Steganography Techniques

Steganography, the practice of concealing information within other data, presents intriguing challenges when applied to grayscale images. This section explores the theoretical foundations and practical implementation of text embedding within PGM images, addressing encoding strategies, capacity limitations, and detection mechanisms.

3.1 Encoding Strategy

The fundamental challenge in image steganography lies in balancing information capacity with imperceptibility. The approach adopted in this implementation utilizes the least significant bits (LSB) of pixel values to store encoded text. Each character is represented using its ASCII value, which requires 7-8 bits of storage. By distributing these bits across multiple pixels, the visual impact of the embedded information can be minimized.

Specifically, each pixel's least significant bit is modified to store one bit of the message. This modification changes the pixel value by at most one unit, which typically remains imperceptible to human observation, particularly in images with moderate to high gray level variation. For a standard ASCII character requiring 8 bits, eight consecutive pixels are needed. This encoding density determines the maximum message capacity based on image dimensions.

3.2 Message Boundary Detection

Determining where the hidden message ends presents a critical design decision. Three primary approaches were considered: fixed-length messages,

explicit length encoding, and sentinel termination. The implementation employs a hybrid strategy, storing the message length in the first several pixels, followed by the message content, and terminated with a null character as an additional safeguard.

This approach provides redundancy against decoding errors while maintaining reasonable efficiency. The message length is encoded using a fixed 16-bit integer, allowing messages up to 65,535 characters (though limited by image capacity). This metadata consumes only two pixels, leaving the vast majority of image capacity available for the actual message content.

3.3 Decoding Process

The decoding algorithm reverses the embedding process, extracting bits from pixel LSBs and reconstructing characters. The procedure begins by reading the message length from the designated header pixels, then extracting the specified number of characters from subsequent pixels. Boundary checking ensures that the decoder does not attempt to read beyond image dimensions or the encoded message length.

Error handling considerations include validation of the decoded length value (ensuring it does not exceed image capacity), verification of the null terminator, and checking for consistent bit patterns. These safeguards help detect corrupted or improperly encoded images, though they cannot guarantee perfect reliability in all scenarios.

3.4 Limitations and Considerations

LSB steganography, while simple and effective, exhibits several limitations. First, the embedded data is vulnerable to lossy image compression, which may alter pixel values and corrupt the hidden message. Second, the technique is susceptible to statistical analysis attacks that can detect anomalies in the least significant bit distribution. Third, any image processing operations (resizing, rotation, filtering) typically destroy the embedded information.

Furthermore, the capacity-imperceptibility trade-off becomes more pronounced in images with low gray level variation or small dimensions. Embedding data in regions with minimal natural texture increases the likelihood of creating detectable patterns. Advanced steganographic techniques address these limitations through adaptive embedding, encryption, and error correction codes, though such enhancements fall beyond the scope of this foundational implementation.

4 Results and Discussion

The implementation was tested extensively using various PGM images with different dimensions, grayscale ranges, and content characteristics. This section presents the findings from these experiments and discusses their implications for practical image processing applications.

4.1 Performance Analysis

The core operations demonstrated consistent and predictable performance characteristics. Memory allocation and deallocation executed in time proportional to image dimensions, as expected for dynamic array operations. The deep copy function exhibited linear time complexity relative to the total number of pixels, making it suitable for real-time applications with moderately sized images.

Rotation operations proved more computationally intensive, particularly for arbitrary angles requiring trigonometric calculations and floating-point arithmetic. The optimized 90-degree rotation functions executed approximately 10-15 times faster than the general rotation function, validating the decision to provide specialized implementations for common transformation angles. For a typical 640×480 pixel image, general rotation completed in approximately 50-80 milliseconds on standard hardware, while 90-degree rotation required only 5-8 milliseconds.

4.2 Steganography Effectiveness

Testing of the steganographic functions revealed both strengths and

limitations of LSB embedding in PGM images. Messages up to several hundred characters were successfully embedded and extracted with perfect fidelity in images containing sufficient visual complexity. The modifications to pixel values remained imperceptible to casual visual inspection, with average pixel changes of precisely one unit as predicted by the theoretical model.

However, embedding in images with large uniform regions (such as solid backgrounds) produced subtle but detectable patterns when examined carefully. This observation confirms the importance of selecting appropriate carrier images for steganographic applications. Images with natural textures, photographic content, or varied gray levels provided superior concealment compared to synthetic images with geometric patterns or limited tonal ranges.

4.3 Edge Cases and Error Handling

The implementation successfully handled various edge cases and error conditions. Boundary checking prevented out-of-bounds access attempts, returning appropriate error codes rather than causing undefined behavior. Memory allocation failures were detected and reported gracefully, allowing calling code to respond appropriately rather than proceeding with invalid pointers.

Attempting to embed messages exceeding image capacity triggered appropriate warnings and truncation behavior. The decoding function demonstrated robustness when processing images without embedded messages, returning empty results rather than interpreting random pixel patterns as meaningful data. These behaviors indicate that the error handling

mechanisms provide reasonable safeguards for practical use.

5 Conclusion

This project successfully demonstrated the implementation of fundamental image processing operations for PGM format files using C programming. The developed library provides a solid foundation for manipulating grayscale images through geometric transformations, intensity operations, and steganographic techniques.

5.1 Key Achievements

The project achieved its primary objectives of creating a comprehensive image processing framework. The data structure design proved effective for representing images in memory while providing efficient access to pixel data. The implementation of rotation, negation, and steganographic functions demonstrated practical applications of fundamental programming concepts including dynamic memory management, algorithmic design, and bitwise operations.

Particularly noteworthy was the successful implementation of LSB steganography, which effectively concealed textual information within images while maintaining visual fidelity. The encoding and decoding mechanisms proved reliable across a range of test images and message lengths, demonstrating the viability of simple steganographic techniques for low-security applications.

5.2 Learning Outcomes

Working directly with low-level image representations provided valuable insights into the computational foundations of image processing. The project reinforced understanding of memory management best practices, particularly the importance of proper allocation and deallocation to prevent memory leaks. Implementation of geometric transformations deepened understanding of coordinate systems, matrix operations, and interpolation techniques.

The steganography component introduced concepts relevant to information security and highlighted the trade-offs between capacity, imperceptibility, and robustness that characterize covert communication systems. These lessons extend beyond the specific context of PGM images to broader principles of data hiding and digital watermarking.

5.3 Future Enhancements

Several potential enhancements could extend the capabilities of this implementation. Support for other Netpbm formats (PBM for binary images, PPM for color) would broaden applicability. Implementation of advanced interpolation methods (bilinear, bicubic) would improve the quality of rotation and scaling operations. Addition of filtering operations (blur, sharpen, edge detection) would provide more comprehensive image processing capabilities.

In the steganographic domain, implementing encryption of embedded messages would enhance security, while error correction codes would improve robustness against image modifications. Adaptive embedding strategies that

select embedding locations based on local image complexity could improve imperceptibility. Statistical analysis tools could evaluate the detectability of embedded messages.

5.4 Final Remarks

This project successfully bridged theoretical computer science concepts with practical implementation challenges. The resulting C library demonstrates that fundamental image processing operations can be implemented efficiently using standard programming techniques and data structures. The experience gained from working at this level of abstraction provides a strong foundation for understanding higher-level image processing frameworks and computer vision applications. The principles explored here—efficient memory management, algorithmic problem-solving, and attention to edge cases—represent transferable skills applicable across diverse software engineering domains.

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